

The Editors
Computational Economics
Springer

Re: Revised submission — **“Online Anchor-Robust Forecasting under Economic Regime Shift”**

Dear Editors,

Thank you for the opportunity to submit a revised version of our manuscript, and please convey our thanks to the reviewer for an exceptionally constructive sequence of reports. The successive reviews have materially improved the paper: they pushed us from an over-claimed framing toward an honest, carefully scoped contribution, and they prompted two new experiments that we believe close the last substantive gaps.

We summarise the paper’s positioning, then respond point by point to the most recent report. All changes are in the resubmitted manuscript; line and section references below are to that version.

Summary of the contribution

We present OARF, an online (streaming, leak-free) implementation of anchor regression for forecasting under economic regime shift, together with an interventional-robustness evaluation protocol and a diagnostic empirical study. We are deliberate about scope:

- The *fixed-channel* method is the core contribution. It carries a fixed-anchor interventional-regret guarantee (two rates, $O(\sqrt{T(1+K)})$ with oracle moments and $O((1+K)T^{2/3})$ with online EMA moments) and, on a controlled SCM with a held-out intervention grid, cuts worst-case interventional error well below reactive and distributionally robust online learners.
- The *learned-channel* add-on (OARF-CD) is presented as exploratory and diagnostic only: it is heuristic, unstable across seeds, and not covered by the theorem.
- On sixteen years of public daily energy variance-proxy data the method is competitive but *not* statistically superior — every retained method lies in the Model Confidence Set — and we report this null result plainly.

The paper is thus a methodological and diagnostic contribution: a streaming anchor-regression update, an interventional-robustness protocol, and stress tests and scenarios mapping when proactive anchor robustness helps and when it fails.

Point-by-point response to the latest report

1. Keyword “causal robustness” may invite stricter causal-identification expectations.

Response. We replaced it with “anchor robustness; interventional robustness” and use “anchor-based interventional robustness” consistently in the text.

2. Related work still says OARF “learns the anchor channel from data.”

Response. Revised to: "...makes the estimator online; we additionally study a heuristic add-on that attempts to estimate the anchor channel from data, while treating the analyst-supplied channel as the primary mode."

3. "OARF aims to make the trigger unnecessary" is too strong.

Response. Softened to: "...tests whether, when an informative anchor is available, part of the need for reactive triggering can be reduced ..."

4. "Never dominated" reads as a formal Pareto claim.

Response. We removed dominance language. The real-data statement now reads: "OARF is statistically indistinguishable from the strongest baselines and broadly competitive across metrics, but the public daily data do not provide statistically decisive evidence of superior performance for any method, OARF included."

5. The contribution may now sound too modest; sharpen it.

Response. We added an explicit "What to cite this paper for" statement to the conclusion, naming the three publishable contributions: (i) a principled streaming form of anchor regression with a fixed-anchor interventional-regret guarantee; (ii) an interventional-robustness evaluation protocol; (iii) a diagnostic study, with a pre-deployment checklist, for deciding when proactive anchor robustness is appropriate.

6. The Ader/AdaGrad mismatch remains a theoretical weakness; verify the claim or do not make it.

Response. We took this seriously and *tested* it rather than asserting it. We implemented the Ader-wrapped variant (a five-expert step-size pool with a Hedge meta-learner and projected OGD) and compared it with the deployed AdaGrad model on the synthetic benchmark over 12 seeds (new Appendix B). Both step rules deliver the interventional-robustness benefit; their per-seed worst-case errors are rank-correlated at Spearman $\rho = 0.80$; and AdaGrad is in fact slightly more accurate and stable. The theory/implementation remark now cites these measured numbers.

7. Visually separate the learned-channel method.

Response. OARF-CD is now labelled "exploratory" in the synthetic results table and the baseline table, and is described throughout as an add-on rather than a co-equal method.

8. Add a "What to cite this paper for" paragraph.

Response. Added to the conclusion (see point 5).

9. Box the practical recipe.

Response. The diagnostic recipe is now a boxed "Pre-deployment checklist for OARF" with the four checks (domain-valid anchor; stable anchor-residual correlation; warm-up stress-test benefit; fallback if the channel is unstable), plus an explicit gating/fallback rule.

10. More careful language around "source of shifts."

Response. We now describe the macro-financial series as an "observable shift channel" or co-moving proxy and reserve the strict causal reading for the synthetic SCM, where it holds by construction.

11. The real-data event study is still absent.

Response. Added (new Appendix C, with figure). We run fixed-anchor OARF and the OGD floor around the COVID-19 crash and the 2022 energy shock, plotting the cumulative squared-loss gap and the anchor-residual correlation. We report the result honestly: OARF holds the anchor-residual correlation below OGD in *every* window (the mechanism is active), but leads on loss in only one of four window/commodity combinations — consistent with, and concretising, the null aggregate result rather than demonstrating superiority.

12. Check appendix details for anonymity.

Response. The code repository is public and *not* anonymised; we state this explicitly and offer to provide an anonymised mirror and withhold the link until acceptance should the journal require double-blind review.

13. Final proofreading; check accents, hats, arrows, line breaks.

Response. We diagnosed and fixed the underlying cause of the earlier garbled-glyph rendering: the manuscript had been compiled with T1 font encoding without the Latin Modern / cm-super Type-1 fonts installed, which forced a bitmap Type-3 fallback for the body text. The text is now set in vector Computer Modern Type-1 fonts (verified at the font level), and we re-checked mathematical accents, hats, arrows and long-URL line breaks.

Earlier rounds

For completeness, earlier revisions also: restructured the abstract into problem/method/theory/evidence; rewrote the theorem as “Fixed-anchor OARF interventional regret” with a complete proof; clarified Algorithm 1 (intercept vs coefficient gradients, dimensions, ridge sign, projection); added stress tests and assumption-violation scenarios mapping where the method fails; added warm-up anchor-selector baselines; moved to 50-seed medians with interquartile ranges and an alignment-versus-robustness scatter; and reframed the real-data section around the Model Confidence Set with exact inclusion p -values.

We hope the reviewer finds the manuscript substantially strengthened and now suitable for publication. We are grateful for the care that went into the reports and remain happy to make further adjustments.

Sincerely,
The Authors